

Orientation and Planning of Buildings

Introduction:

- Building orientation refers to the positioning of a building with respect to sun, wind and surroundings.
- Building planning focuses on arrangement of spaces for comfort, efficiency, and aesthetics.
- Proper orientation and planning improve energy efficiency and living comfort.

Objectives of Building Orientation:

- Maximize natural light and ventilation
- Reduce heat gain and energy consumption
- Provide thermal comfort
- Improve health and productivity of occupants

Factors Affecting Orientation:

- Climate and weather conditions
- Sun path and solar radiation
- Direction of prevailing wind
- Topography and surrounding buildings
- Road access and site conditions

Importance of Proper Orientation:

- Better indoor comfort
- Energy savings
- Improved ventilation
- Reduced environmental impact
- Enhanced building durability

Factors to be considered for planning orientation of building are:

(a) To Sun Exposures of Building:

- Calculate Position of Sun Relative to Particular Places, Seasons, Times of Day: Sun's position at any location on earth, at any day or hour of a day may be determined from sun path diagram with its angle of incidence and azimuth angle.
- Orientation of Windows: Orientation, size and shape of individual windows may require checking by reference to sun path diagrams to determine sunlight penetration and effectiveness of required sun controls, extensive and continuous shadowing of ground due to large and tall buildings.

(b) Air Flows Through Buildings:

- The greatest pressure on the windward side of a building is generated when elevation is right angles to the wind direction. So it seems to be obvious that the greatest indoor air velocity will be achieved in this case.

(c) Sensory Factors:

- Views: The consideration of desirable and undesirable view will help to determine the form and orientation of building, the building's fenestration (door and window openings), and the types of vegetation used in landscaping.
- Sound: The level, quality and sources of sound affects the distribution and orientation of the building mass, the choice of building materials and their construction.

Building Planning:

a) Site Planning

- (i) Energy conservation.
- (ii) Passive space heating modes.
- (iii) Active space heating

b) Building Planning

- Planning is the arrangement of rooms and spaces for proper functioning.
- Good planning ensures convenience, safety, privacy, and aesthetics.

Planning of building depends on:

- Functional object and requirements.
- Component parts, size and interrelationship.
- Topography and shape of plot.
- Climatic condition.
- Location and neighborhood.

Principles of Building Planning:

- Aspect: Placement of rooms according to sunlight and air flow.
- Prospect: Attractive view from the building.
- Privacy: Maintain privacy between public and private spaces.
- Grouping: Group related rooms together for convenience
- Circulation: Provide easy movement within the building.
- Flexibility: Allow future modification and expansion.
- Ventilation: Should have enough opening for ventilation.
- Economy: Building should have minimum floor area with maximum utility.
- Sanitation: Provide sufficient light, ventilation and water for cleaning and sanitary conveniences.

Setting out of Building:

Layout Plan of Excavation:

- Before commencement of the excavation of trenches for foundation a layout plan is prepared.
- It is the process of the building plan measurement to the construction site and fixing of excavation line of all the walls, pillars of the plan of the proposed building is called site layout work.

Process of Setting Layout Plan:

- Site clearance work is done ie, removal of debris; plants and other any kind of obstruction and approximately levels the ground.
- Site plan and detailed foundation trench plan with the depth of excavation and other details are noted as per paper work.
- Make the center line of the longest outer wall of building by stretching a string between wooden pegs driven into its ends which is then used a reference line marking the center lines of other walls of the building.
- Center line of the wall which is perpendicular to the wall is marked by setting up a right angle.

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- Pegs are driven into all control points of the wall center lines.
- Mark each corner of the building with suitable measurements.
- Check by measuring opposite lengths of corners diagonally if both of the length equals then setting is ok.

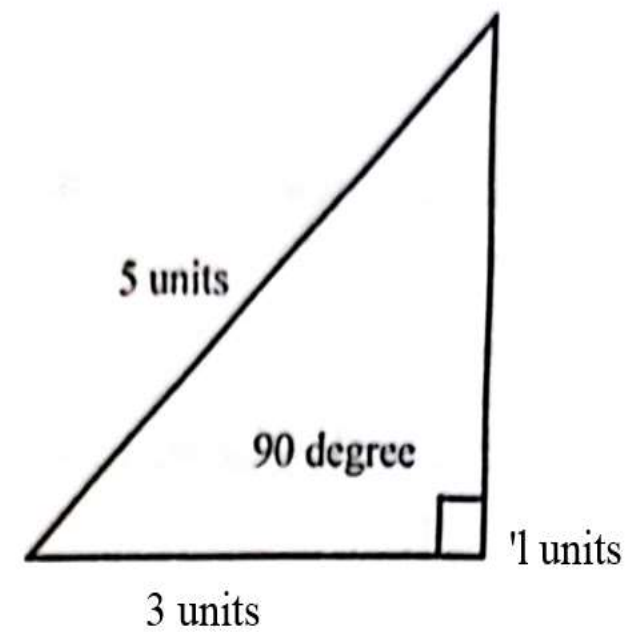
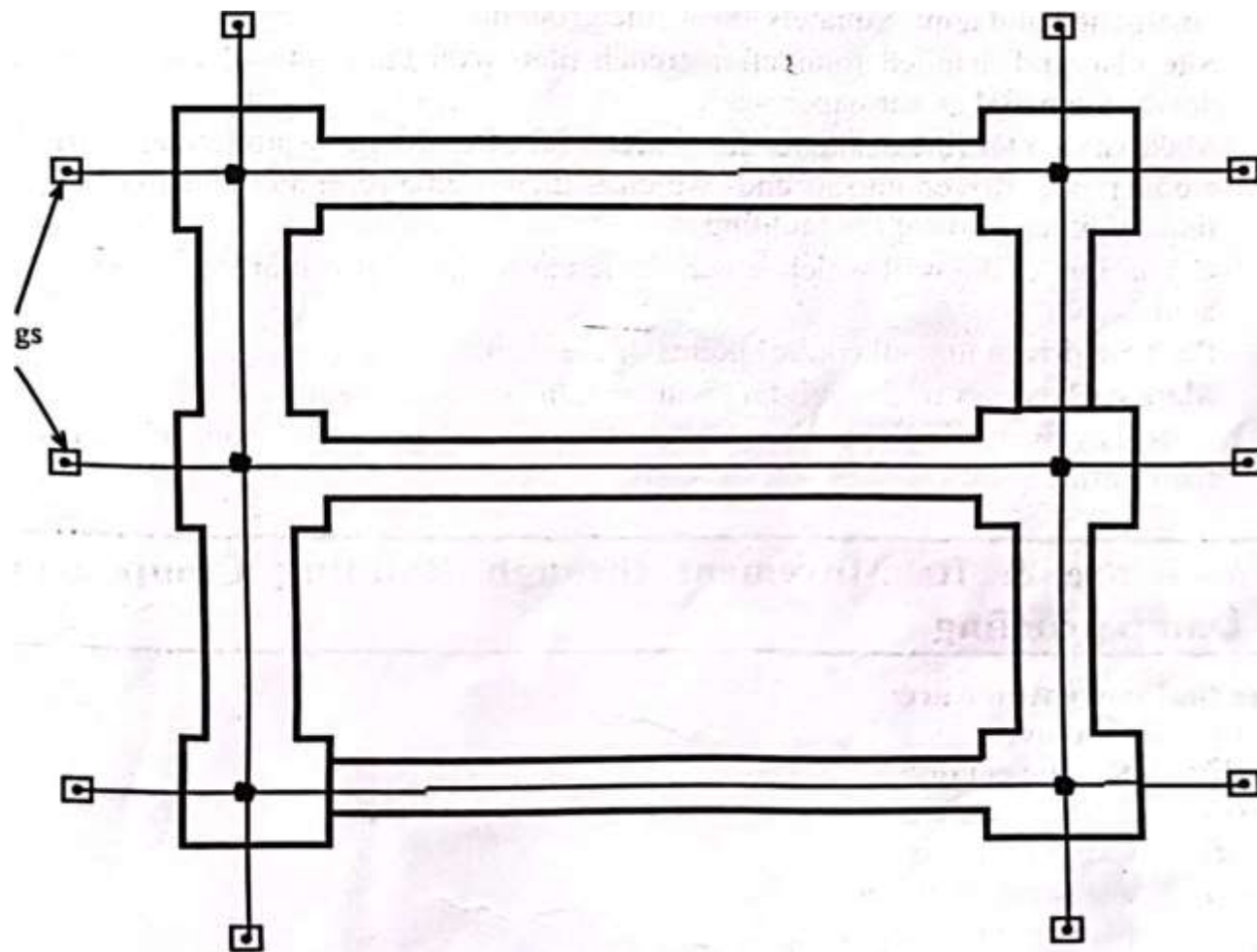


Fig. I. 7: 3-4-5 Rule for Setting Corners.

Lighting and thermal performance:

Lighting:

- Lighting inside a building is necessary to illuminate the interior & its contents to perform the tasks.
- As per NBC 206:2003) rooms used for living, sleeping, studying, eating and working excluding bathrooms, stores, passage etc. shall have admission of light through external wall opening not less than $1/10^{\text{th}}$ of the floor area of the room.

Lightening Requirements:

- Source of light should be steady.
- Degree of level of illumination must be suitable.
- Should not cause excessive heat.

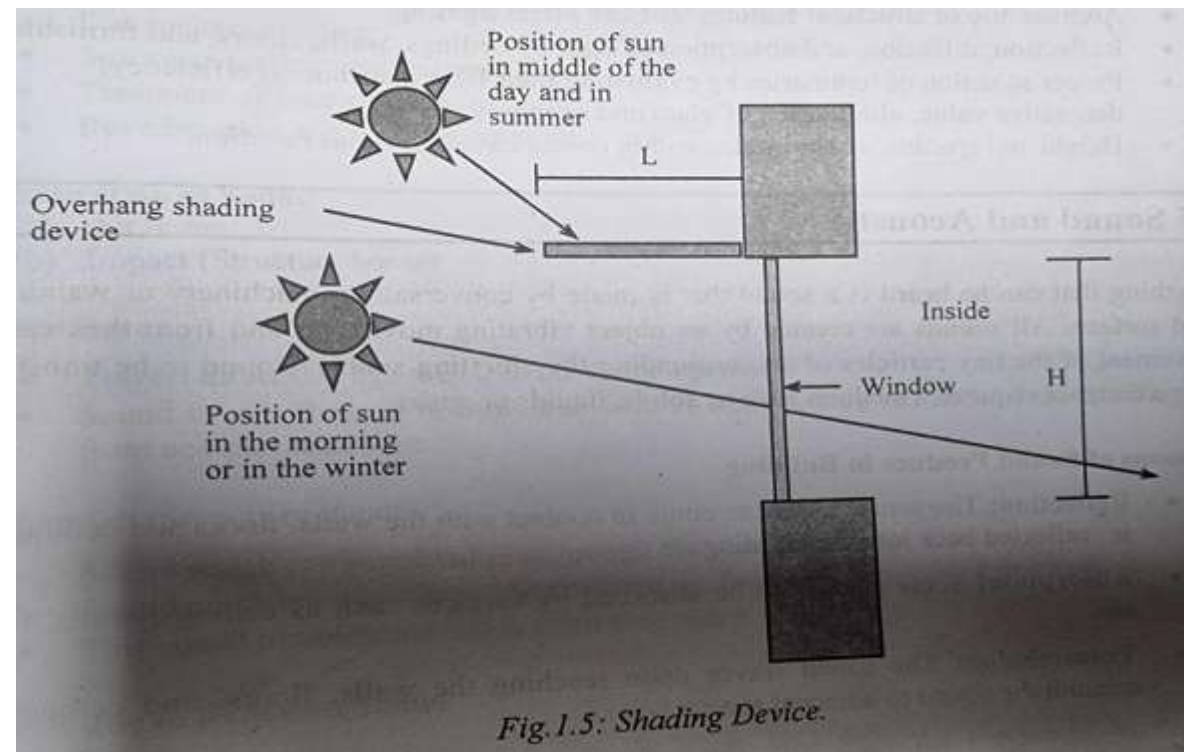
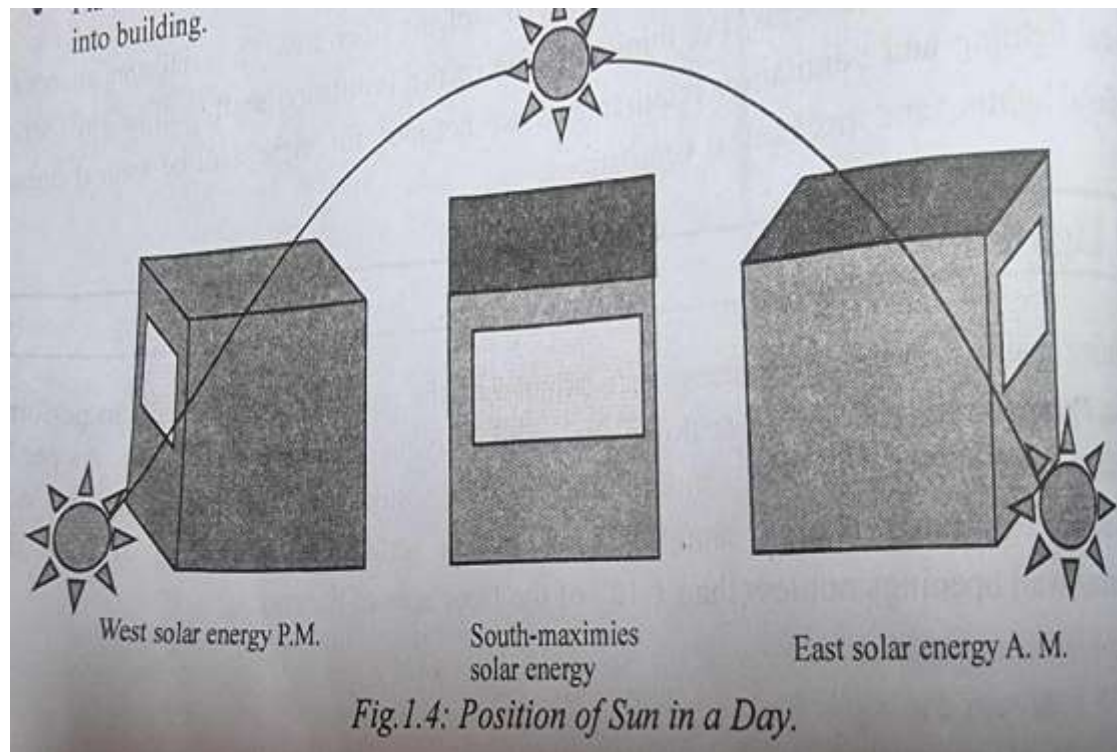
Day light:

- Depth, shape & spacing of building affects the daylight penetration.
- The quantity of daylight that is admitted through a window/roof is expressed as a daylight factor.
- Large glazing areas may cause discomfort due to glare or overheating.
- Careful design consideration are required.

Types of lighting:

a. Natural lighting :

- Source of natural lighting is sun.
- Placement, size, type & orientation of openings are important for penetration of light into building.
- Poorly designed glazing, admitting too much solar radiation at the wrong time of a day or period of the year leads to overheating.
- It can lead to increased energy consumption, need for mechanical cooling.
- Shading device can protect the summer sun and also the monsoon rain.



b) Artificial lighting:

- Light is generated artificially.
- Purpose of artificial lighting system is to provide illumination to perform visual task.
- Quality of lighting varies according to the light diffusion, color & brightness ratio.

Good lighting system:

The principle factors to be considered for satisfactory room lighting are:

- Room dimension: ceiling height & floor area.
- Architecture or structural features that can affect lighting.
- Reflection, diffusion, & absorption of light by ceiling, walls & furnishings.
- Height and spacing of luminaries within rooms based on room function.

Thermal comfort:

- That condition of mind which expresses satisfaction with the thermal environment.
- Keeping the temp., humidity, airflow and radiation sources within acceptable range.
- Maintaining a person's thermal comfort means ensuring that they don't feel too cold and too hot.

Thermal controls can be made through:

1. Passive solar design:

- Basically architecture related with efficient use of solar energy either directly or indirectly to achieve comfort by using simple technologies and architectural components on site.
- By using simple techniques such as orientation, planning, material, technology, vegetation, color of building.

While designing the building according to passive solar design, following factors are taken into account:

Orientation, ventilation (opening), insulation, site planning, building form, space planning, material and technology, shading devices, vegetation, color, texture etc.

Objectives:

- To provide better and thermal performances, natural cross ventilation, day lighting, proper insulation, use of adequate shading devices.
- To reduce energy consumption for cooling, heating and lighting.
- To design and construct to produce comfortable, energy efficient and environment friendly building.

2. Mechanical controls:

- Heating
- Air- conditioning
- Forced ventilation

Ventilation and air conditioning:

Introduction:

- Replacement of stale air by fresh air.
- Predominant contributor to thermal and hygienic environment in building.
- Natural ventilation has great importance for human comfort.
- The satisfactory ventilation should provide a refreshing, healthy and comfortable environment.

Necessary of ventilation:

- To replace stale air by fresh air.
- To prevent undue concentration of bacteria-carrying particles.
- To create healthy living conditions by preventing the undue accumulation of carbondioxide.

Function of ventilation:

- Supply of fresh air
- Convective cooling
- Physiological cooling

Ventilation System:

1. Natural Ventilation System:

Supply of outside air into the building through windows or other openings due to wind outside and convection effects arising from temp. or vapor pressure differences of both between inside or outside of the building.

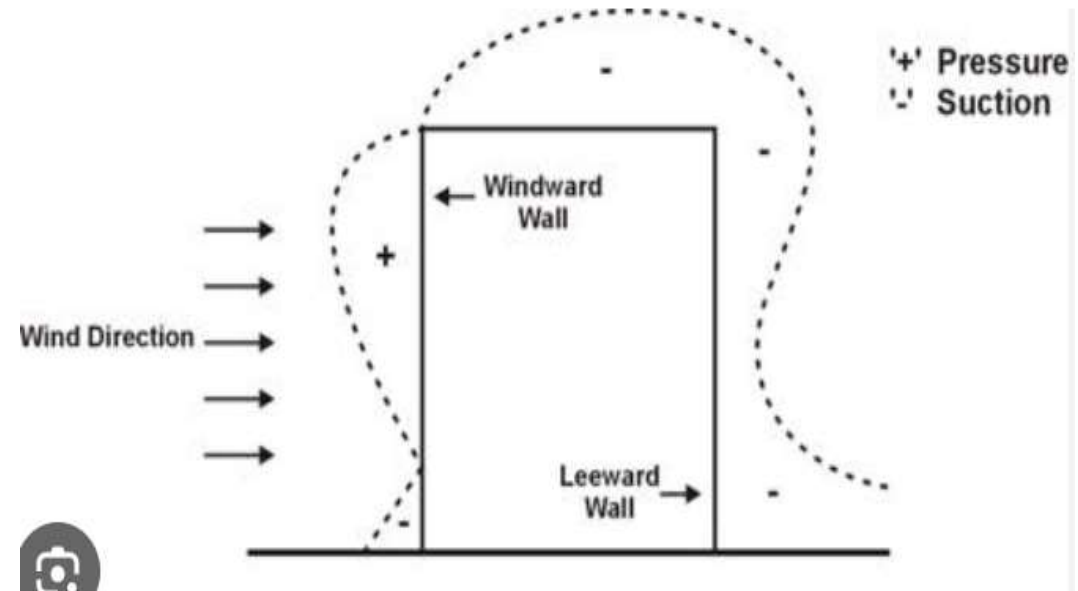
Natural Ventilation Rules:

- Building and ventilating equipment should be oriented towards a particular wind direction.
- Inlet opening should not be obstructed by buildings, trees, signboards etc.
- For temp. differences there must be vertical distance between openings and vertical distance should be great as possible.
- The opening should be accessible to and operable by occupants.

Natural Ventilation is Achieved by Following Mechanism:

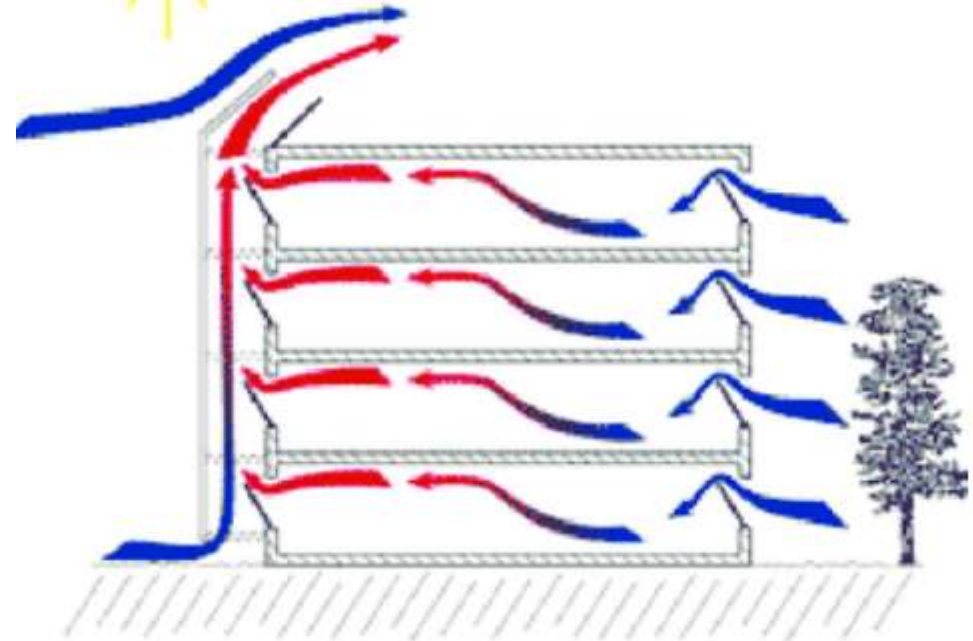
a) Wind Driven Ventilation:

- Wind-driven ventilation, also known as cross-ventilation or wind effect ventilation, is a passive cooling technique that uses wind to naturally ventilate a building by drawing in fresh air through openings on the windward side and expelling stale air through openings on the leeward side.



b) Stack Ventilation:

- Stack ventilation uses temperature differences to move air. Hot air rises because it is lower pressure. For this reason, it is sometimes called buoyancy ventilation.
- The stack effect: hot air rises due to buoyancy, and its low pressure sucks in fresh air from outside.



c) Ventilation Due to Combined Effect of Wind and Stack:

- Ventilation utilizing both wind and stack effects, also known as the "chimney effect," involves using natural pressure differences created by wind and temperature differences to draw fresh air into and exhaust stale air out of a building.

2) Mechanical Ventilation:

- It is the mechanism of supply outside air, either by positive ventilation or by infiltration by reducing inside pressure due to exhaust air, or by a combination of positive ventilation & exhaust of air.

Sound and Acoustic:

- Any thing that can be heard is a sound.
- That can be made by conversation, machinery etc.
- Sound to be transferred over a distance requires a medium such as solid, liquid or gases.

Reasons of sound produce in building:

- Reflection
- Absorption
- transmission

Noise:

It is defined as the unwanted sound & have the following effects:

- Irritation
- Disturbance of sleep
- Disturbance to hold a normal conversation
- Damage the ear

Acoustic:

- The science of sound which deals with origin, propagation & auditory sensation of sound.
- It assure the optimum condition for producing & listening to speech, music etc.
- Sound proof & insulation against noise is essential for film studio, TV & radio station, hospitals etc.

General principle of acoustical design:

Some of planning principles & factors which plays important role in creating acoustical condition are as follows:

- Site selection & planning
- Treatment of interior surface
- Shape of room of building
- Dimensions of room
- Seats & seating arrangements

Classification of sound:

- a) Air borne
- b) Impact (structure borne)

(a) Air borne:

- Travels direct through the air to the ear of person.
- Sound travels through open doors/ windows from outside to inside of the building and from one part to another.

(b) Impact (structure borne):

- Sound transmitted first through the structure.
- Sound produced direct from the structure (footsteps, vibration, hammering)

Defects due to reflected sound:

a) Echoes:

- Echoes are produced sound waves after reflection from the surface of walls, roofs, ceiling etc.
- Repeated sound coming from the same sources.
- Direct sound wave is already heard & thus there is repetition.

b) Reverberation:

- Reverberation is due to reflection in an enclosed space.
- Generally noticed in public halls, auditoriums etc.

Solution to noise problem:

- Devices are mufflers & mounting machinery on resilient pad.
- Provide reasonable degree of sound insulation.
- Isolate the source & receiver.

Sound insulation:

It is the prevention of transmission of sound.

Methods of sound insulation:

a) Constructional measures:

- Partition of porous materials construction
- Hollow & composite wall construction
- Use of resilient surface materials on floors
- Floating floor construction

b) Insulation of windows & doors:

- Double panes of glass with 150-200 mm air space.

c) Insulation of machinery :

- Mount machinery by resilient materials
- Insulate generators

d) Insulation of sanitary fittings:

- Mount water closet on resilient materials
- Wrap sanitary pipelines

e) Use of sound absorbent:

- Percentage of sound absorbed by materials
- Materials which are flexible

Types of sound absorbent:

1) Porous absorbers:

- Porous absorbers are the most commonly used sound absorbing materials.
- Common porous absorbers include carpet, plaster, wood wool glass fiber etc.

2) Resonant absorbents:

- Absorbent materials is fixed on timber framing with an air space left between the framing & wall.
- Most efficient for absorbing sound waves at low frequency.

3) Cavity resonators:

- Consists of a chamber with a narrow opening in which absorption takes place.
- Such arrangements is useful for selected frequency from individual machines, air conditioning plants etc.

4) Composite type of absorbents:

- Panels may be of metal, wood, hard boards etc
- Whole in the panel vary between 10-20% of total area of panel

Thank You